

IoT Deliverable One

Group Members

Donell Bikketi- 166604

Gloria Kendi- 166074

Emmanuel Douglas- 168053

Crystal Kanana- 169820

Andrew Kigonde- 166228

Q1) Environmental Requirements for Carnations

Parameter	Optimal Range/Value	Notes
Temperature	10–24°C (50–75°F)	Ideal daytime temperature is 18–24°C; night temperature is 10–11°C in winter and 13–15.5°C in summer. Temperatures consistently above 30°C stress the plant, while below 8–9°C halts flowering
Relative Humidity	60–85%	Humidity should be maintained at 80–85% during the beginning of vegetative growth, and 60–65% during the full growth stage
Soil Type	Sandy loam/loamy	Sandy loam soils rich in organic matter are considered most suitable for carnation production, as they provide good drainage while retaining adequate moisture.
Soil Moisture Content	21-40%	The majority of flowers require soil moisture levels between 21% and 40%. Carnations specifically prefer the soil to dry slightly between waterings to avoid root rot.
Soil pH	6.7-7.0	Carnations should be planted in well-drained soil that is fertile and slightly alkalized, with a pH of 6.7-6.9

Sunlight Exposure	4-6 hours/day	Carnations at least 4-6 hours of direct sunlight daily. At least hours is ideal; morning sunlight is especially beneficial as it helps keep flowers fresh and prevents color fading.
-------------------	---------------	--

Q2) Hardware Components

Microcontroller & Communication

Component	Purpose
ESP32S DevKIT WiFi + BLE Module (30-pin)	Main microcontroller — processing, WiFi/BLE communication

Sensors

Component	Purpose
DHT22 AM2302	Temperature & relative humidity sensing
MQ-5 Gas Sensor	LPG / methane / butane / propane detection
Capacitive Soil Moisture Sensor (e.g. v1.2)	Soil moisture content measurement
Soil pH Sensor Module	Soil pH measurement
LDR (Light Dependent Resistor) / BH1750 Light Sensor	Sunlight exposure measurement

Output & Control

Component	Purpose
1.3" White IIC 128×64 OLED LCD Display	Displaying sensor readings locally
5V 1-Channel Low-Level Trigger Relay Module	Actuator control (e.g., triggering fans, pumps, alerts)

Power Supply

Component	Purpose
5V Micro USB Power Supply / Power Bank	Powering the ESP32 and peripherals
3.3V LDO Voltage Regulator (e.g., AMS1117-3.3)	Stepping down the voltage for sensitive sensors

Passive / Discrete Components

Component	Purpose
10kΩ Resistors (×5)	Pull-up/pull-down resistors for sensors (e.g., One DHT22 data line)

1kΩ Resistors (×3)	Current limiting resistors
10kΩ Potentiometer	Analog signal calibration/voltage divider
100nF Ceramic Capacitors (×4)	Decoupling/noise filtering near sensors
10μF Electrolytic Capacitors (×2)	Power supply smoothing

Prototyping Tools

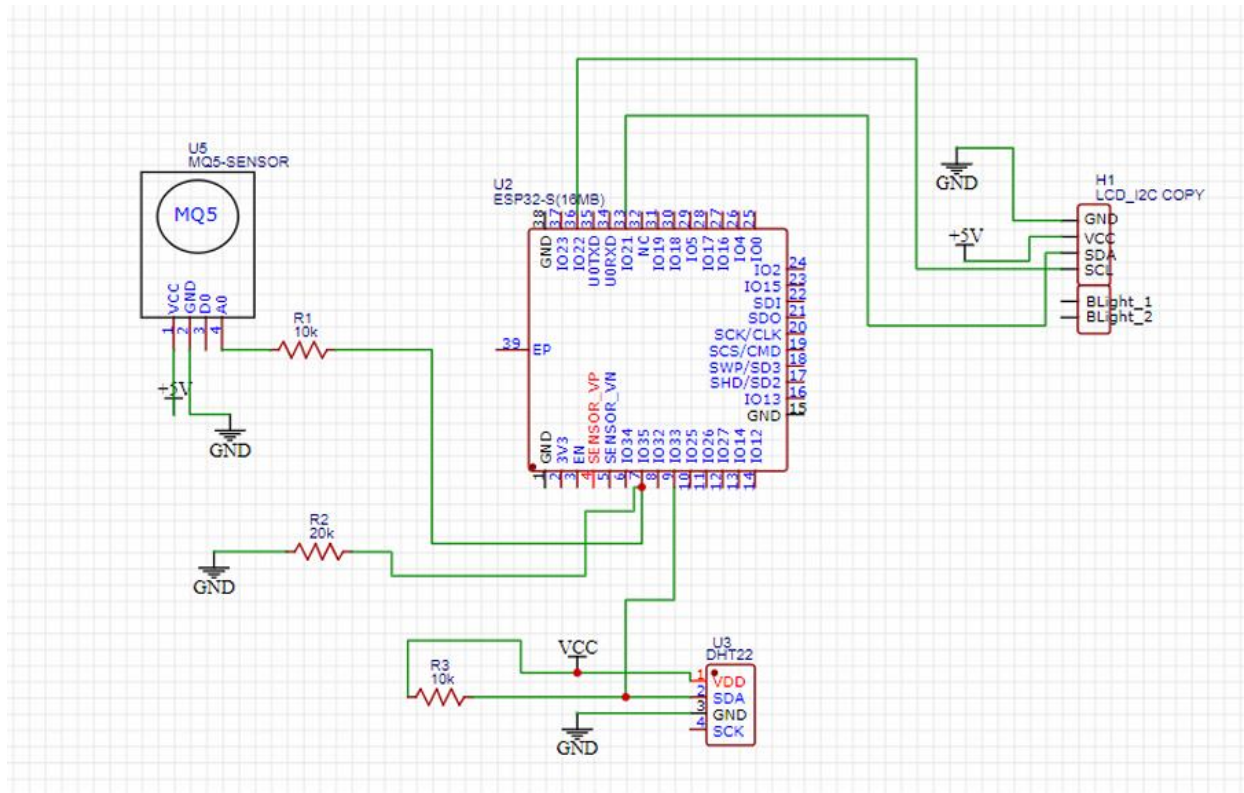
Component	Purpose
Full-size Breadboard (×2)	Prototyping circuit connections without soldering
Jumper Wires (M-M, M-F, F-F assortment)	Connecting components on the breadboard
USB to Micro-USB Cable	Uploading code to ESP32 from PC
Multimeter	Testing voltages, continuity, and debugging
Laptop / PC with Arduino IDE or VS Code + PlatformIO	Programming and flashing the ESP32

Q3) Component Datasheets & Reference Links

Component	Datasheet/ Reference Link
1.3" White IIC 128×64 OLED LCD (SH1106 controller)	Product page + datasheet: https://www.displaymodule.com/products/1-3-inch-oled-graphic-display-128x64-with-i2c — links to both the Product Datasheet and SH1106 Controller Documentation
ESP32S DevKIT WiFi + BLE Module (30-pin)	Official Espressif datasheet (PDF): https://www.espressif.com/sites/default/files/documentation/esp32_datasheet_en.pdf
DHT22 AM2302 Temperature & Humidity Sensor	Official Adafruit-hosted datasheet (PDF): https://cdn-shop.adafruit.com/datasheets/Digital+humidity+and+temperature+sensor+AM2302.pdf
MQ-5 LPG, Natural Gas, Coal Gas Sensor	Official Winsen manufacturer datasheet (PDF): https://www.winsen-sensor.com/d/files/MQ-5.pdf
5V 1-Channel Low-Level Trigger Relay Module	Reference datasheet (PDF): https://handsontec.com/dataspecs/relay/1Ch-relay.pdf

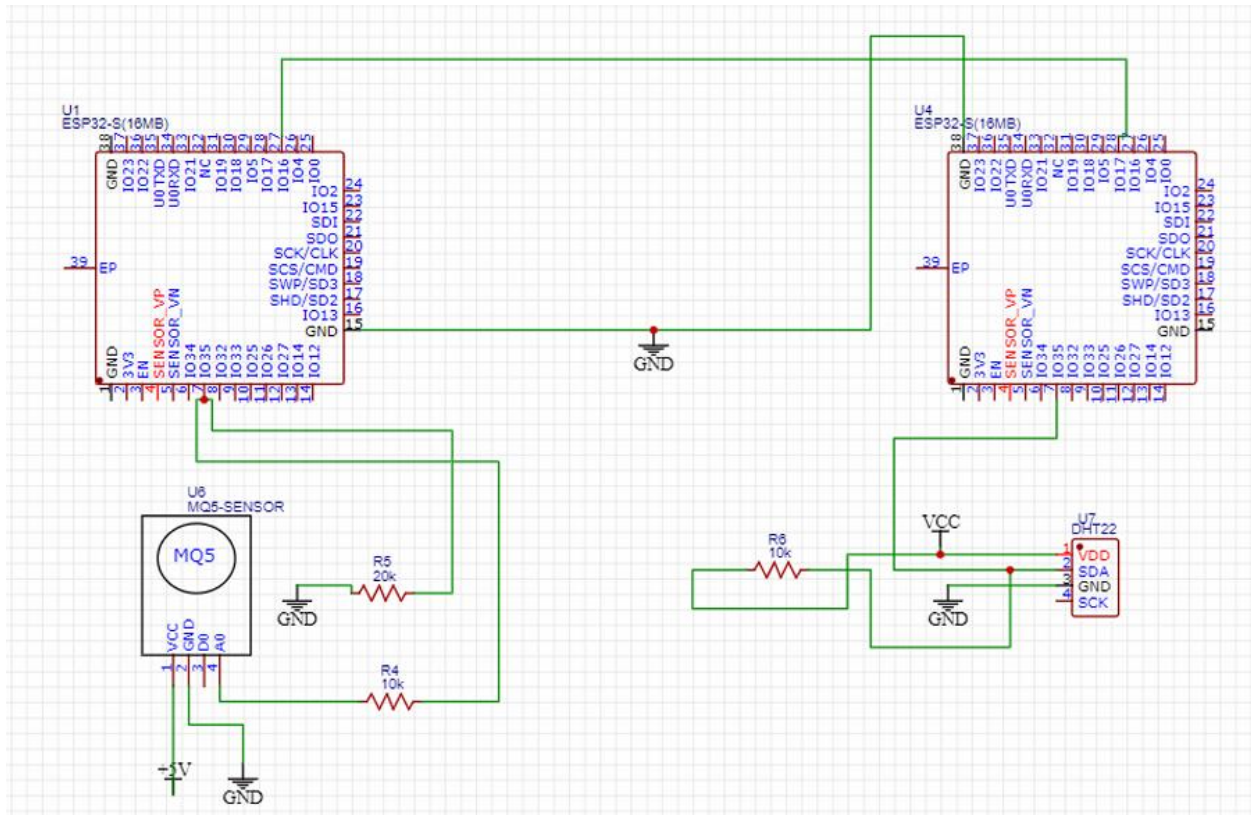
Circuit 1

1 ESP32S connected to 1 MQ-5, 1 DHT22 and 1 LCD



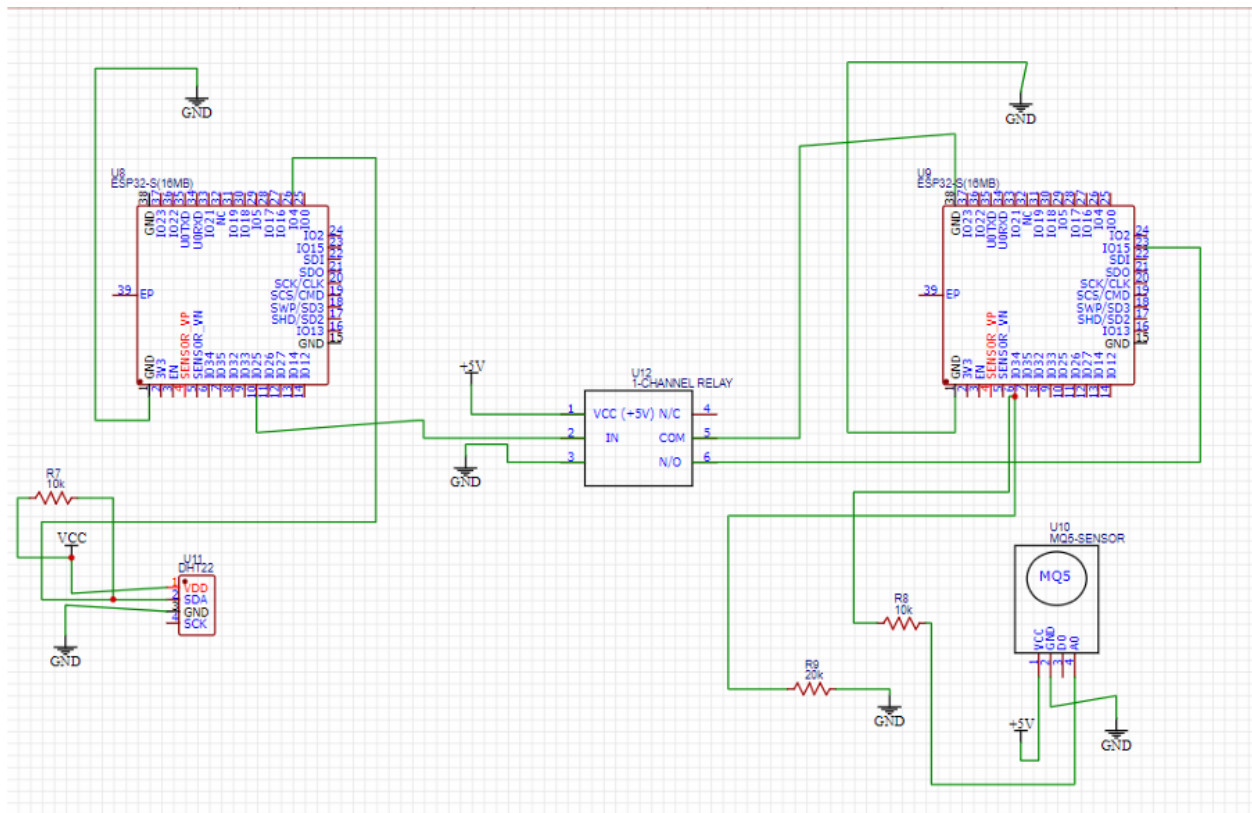
Circuit2

1 ESP32S connected to 1 MQ-5 interfaced directly with another ESP32S connected to 1 DHT22



Circuit 3

1 ESP32S connected to 1 DHT22 connected to 1 relay which is connected to another ESP32S connected to 1 MQ-5



Evidence of groupwork.

